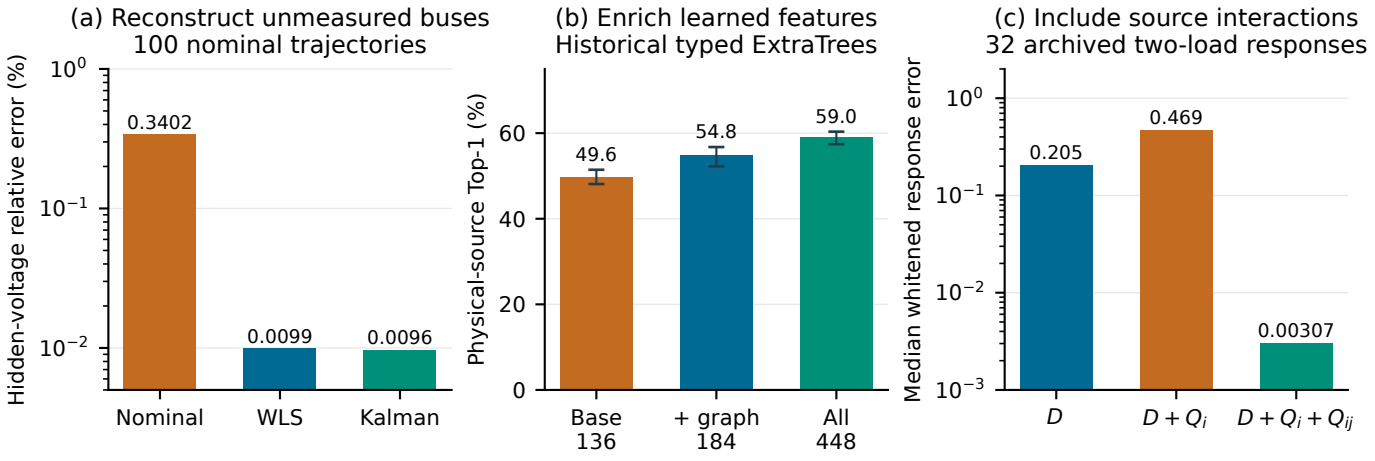
(a) Reconstruct unmeasured buses
100 nominal trajectories
Hidden-voltage relative error (%)
10^{0}
10^{-1}
10^{-2}
0.3402
0.0099
0.0096
Nominal
WLS
Kalman
(b) Enrich learned features
Historical typed ExtraTrees
Physical-source Top-1 (%)
60
40
20
0
49.6
54.8
59.0
Base
136
+ graph
184
All
448
(c) Include source interactions
32 archived two-load responses
Median whitened response error
10^{0}
10^{-1}
10^{-2}
10^{-3}
0.205
0.469
0.00307
D
$D + Q_i$
$D + Q_i + Q_{ij}$